

Thyroid Diseases (Dr Sarmad)

The Big Picture

Butterfly gland, front of neck, **10–20 g**, makes **T4 + T3** under **TSH** (anterior pituitary) control via negative feedback; **TRH** from hypothalamus drives TSH. **Calcitonin** (parafollicular C-cells) lowers calcium. Master the **hyper- vs hypothyroid** picture, the TFT pattern, the **thyroid storm** emergency, and **thyroidectomy complications**.

Axis & Hormones

- Thyroid → T4 + T3; controlled by **TSH** (negative feedback).
- **Hyperthyroid** → **TSH suppressed**; **hypothyroid** → **TSH stimulated**.
- Calcitonin (C-cells) — stimulated by hypercalcaemia, lowers serum Ca.

Hyperthyroidism

- **Symptoms:** weight loss, ↑ appetite, anxiety, irritability, heat intolerance, **tachycardia**, tremor; **Graves = eye signs**.
- **Causes (primary):** **Graves disease**, toxic multinodular goitre, toxic adenoma.
- **Graves ophthalmopathy:** eyelid retraction, **proptosis**, periorbital oedema, chemosis (orbital fat/muscle inflammation).
- **Secondary:** pituitary adenoma (TSH-secreting).
- **Without hyperfunction:** subacute thyroiditis (leak), iatrogenic overdose.

Hypothyroidism

- **Symptoms:** fatigue, ↓ appetite, **weight gain**, **constipation**, **cold intolerance**, **dry skin**.
- **Primary atrophic:** **Hashimoto** (autoimmune), end-stage Graves, iatrogenic (**I-131**, **thyroidectomy**, **radiation**), biosynthesis defects, iodine deficiency/excess, agenesis.
- **Secondary:** pituitary (panhypopituitarism, isolated TSH deficiency), hypothalamic.
- **Transient:** silent/subacute thyroiditis, thyroxine withdrawal.
- Early detection in childhood prevents **mental retardation** + oral/skeletal complications.

Thyroiditis (causes)

Acute suppurative, subacute painful, subacute painless, **Hashimoto**, chronic fibrosing (**Riedel**).

Investigations

- **TFT:** T3, T4, TSH.
- **Hyperthyroid:** ↑T3, ↑T4, ↓TSH.
- **Hypothyroid:** ↓T3, ↓T4, ↑TSH.
- Neck **ultrasound**; **FNAC** (cytology of nodules); **thyroid scan** (Tc-99m — toxic adenoma = hot nodule).

Thyroid Storm (Thyrotoxic Crisis) — emergency

- Life-threatening sudden hormone surge complicating hyperthyroidism.
- **Triggers:** untreated/poorly controlled hyperthyroid, stress, infection, trauma, certain drugs.
- **Features:** **high fever**, **tachycardia**, **agitation/confusion**, **vomiting/diarrhoea**, **muscle weakness**, **coma**.
- **Treatment:** admit; **β-blockers**, **antithyroid drugs**, **iodine**, **glucocorticoids**.

Thyroidectomy Complications

- **Hypocalcaemia = most common** (parathyroid injury) → calcium + vitamin D.
- **Recurrent laryngeal nerve** injury → hoarseness (often recovers).
- **Bleeding/haematoma** → airway pressure → **emergency re-operation**.
- Less common: **superior laryngeal nerve** (voice quality/high notes), infection, oesophageal/tracheal injury, thyroid storm, hypothyroidism (if total + no replacement), seroma, surgical emphysema (transoral).

Walk-Away Points

- **Hyper:** ↑T3/T4, ↓TSH, Graves + eye signs. **Hypo:** ↓T3/T4, ↑TSH, Hashimoto.
- **Thyroid storm** = fever + tachycardia + confusion → β-blocker + antithyroid + iodine + steroid.
- Thyroidectomy: **hypocalcaemia commonest**, RLN → hoarseness, **haematoma** → **airway emergency**.

Thyroid Diseases — Revision Table			
Axis	Fact		
Gland	Butterfly, 10–20 g, T4 + T3		
Control	TSH (ant. pituitary), TRH (hypothalamus)		
Calcitonin	C-cells, lowers Ca		
TFT pattern	T3	T4	TSH
Hyperthyroid	↑	↑	↓
Hypothyroid	↓	↓	↑
Hyperthyroidism	Detail		
Symptoms	Weight loss, heat intolerance, tachycardia, tremor		
Primary causes	Graves , toxic MNG, toxic adenoma		
Graves eyes	Lid retraction, proptosis , chemosis		
Secondary	TSH-secreting pituitary adenoma		
Hypothyroidism	Detail		
Symptoms	Fatigue, weight gain, constipation, cold, dry skin		
Primary	Hashimoto , I-131, thyroidectomy, iodine def		
Secondary	Pituitary, hypothalamic		
Thyroiditis	Acute suppurative, subacute painful/painless, Hashimoto, Riedel		
Investigation	Use		
TFT	T3/T4/TSH		
US neck	Structure		
FNAC	Nodule cytology		
Tc-99m scan	Hot toxic adenoma		
Thyroid storm	Detail		
Features	Fever, tachycardia, agitation, coma		
Triggers	Infection, stress, trauma		
Treatment	β-blocker, antithyroid, iodine, glucocorticoid		
Thyroidectomy complication	Note		
Hypocalcaemia	Most common (parathyroid)		
RLN injury	Hoarseness		
Haematoma	Airway emergency → re-op		
Superior laryngeal n.	High notes / voice		

Bold = high-yield.